

# Jasper Vermeulen

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PhD Researcher | Human-Robot Collaboration | Socio-Technical Systems and Human Factors  
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## Keywords

Human-Robot Interaction; Human-Robot Collaboration; Ethnography; CSCW; Human Factors

## Key Skills

- Human-Robot Collaboration (HRC)
- User Experience (UX) Research
- Human Factors & Ergonomics
- Qualitative Research & Analysis Methods
- Public Speaking
- Project Management & Planning
- Stakeholder Engagement & Industry Collaboration
- Academic Writing

## Education

### **PhD in Human Factors of Human-Robot Collaboration**

Queensland University of Technology, Australia | Expected Defence Date September 2026

**Thesis Title:** Understanding the Human(s) in Human-Robot Collaboration: A Socio-Technical Inquiry into Robot-Assisted Surgeries

*Research Focus:* Human factors in multi-actor Human-Robot Collaboration (HRC) within surgical settings, with an emphasis on the gap between how robot-assisted surgery has been designed and how it is performed in practice.

*Key outcome:* Based on three interrelated empirical studies, including video analysis of robot-assisted surgeries, 100+ hours of ethnographic observation in a Brisbane hospital, and interviews with surgical team members across Australian cities, this research reveals a significant gap between the designed and enacted realities of robot-assisted surgery. Using Socio-Technical Grounded Theory (STGT), these findings are translated into a novel, transferable theory of multi-actor HRC, applicable across healthcare, manufacturing, and construction.

### **Post-Master's in Human-Media Interaction**

Keio University, Japan

*Focus: Japanese Language + Emergent technologies in the context of mitigating loneliness.*

**MA in Human-Media Interaction (High Distinction)**

Erasmus University, Netherlands

*Thesis: User acceptance and attitudes towards smart home doorbells.*

**BSc in Media and Communication (High Distinction)**

Erasmus University, Netherlands

*Thesis: User attitudes towards smart home technologies. (Best Thesis Nominee)*

**Other Education**

**CHI2026: Doctoral Consortium Training**

Remote Attendance | Issued April 2026

**HRI2026: HRI Pioneers (Doctoral Consortium Training)**

Remote Attendance | Issued March 2026

**Professional Training for PhDs (2-day)**

Australian Cobotics Centre, Australia | Issued October 2025

**Effective Assessment and Feedback University Teaching**

Queensland University of Technology, Australia | Issued April 2025

**Approaches to Planning and Designing Learning University Teaching**

Queensland University of Technology, Australia | Issued March 2025

**OzCHI2024: Doctoral Consortium Training**

University of Queensland, Australia | Issued November 2024

**Human Factors in Healthcare: Designing Safe Systems Short Course**

Monash University, Australia | Issued October 2024

**Responsible Use of AI in Research Writing**

Queensland University of Technology, Australia | Issued March 2024

**ICH Good Clinical Practice E6 (R2)**

The Global Health Network | Issued February 2024

**Presenting Research Seminars**

Queensland University of Technology, Australia | Issued September 2023

**Research Project Management**

Queensland University of Technology, Australia | Issued July 2023

## Writing Style and Grammar for Researchers

Queensland University of Technology, Australia | Issued June 2023

### Experience

#### PhD Researcher

Australian Cobotics Centre & Queensland University of Technology (QUT)

Brisbane, Australia | July 2023 – Present

- Conducting research on human factors and ergonomics in Human-Robot Collaboration (HRC) within surgical environments.
- Provided actionable insights to Stryker Inc. for Mako-specific product and workflow improvements (details classified).
- Presented research at national and international conferences, engaging both academic and industry audiences.

#### Research Assistant

Construction Architecture and Robotics Lab (CARL) & QUT School of Architecture and Built Environment

Brisbane, Australia | November 2025 – Present

- CARL is a new research initiative at QUT. Under the guidance of A/Prof Tim Rose and Prof Glenda Caldwell, I conducted a systematic scoping review (PRISMA) to investigate the adoption barriers and enablers of HRC in the construction industry.
- Conference Paper conditionally accepted for presentation at the 43<sup>rd</sup> International Symposium on Automation and Robotics in Construction (June 22-26, 2026) titled "Reconceptualising Human-Robot Collaboration in AEC: A PRISMA Scoping Review."
- Journal paper following the PRISMA review under development.

#### Sessional Academic (Tutor)

Queensland University of Technology, QUTYou010, People with Robots Unit

Brisbane, Australia | September 2024 – Present

- Delivering weekly, 2-hour tutorials and supporting student learning in human-robot interaction and design-related topics.
- Providing academic guidance and marking assessments.
- Involved in redesigning curriculum, leading to improved student engagement captured in student survey.

#### Research Assistant

Australian Cobotics Centre & QUT School of Architecture and Built Environment

Brisbane, Australia | November 2024 – November 2025

- Contributed to the "Brain-Computer Interface (BCI) for HRC" project in collaboration with Dr Alan Burden, exploring how BCI can support humans engaging in various HRC tasks.
- Project outcomes and deliverables classified.

### **Research Assistant**

QUT Centre for Robotics & QUT Design Lab

Brisbane, Australia | January 2024 – November 2024

- Contributed to the “Robotic Blended Sonification” project in collaboration with A/Prof Stine Johansen, exploring how sound design can enhance human-robot collaboration.
- Assisted with study design, participant recruitment, and qualitative data collection and analysis.
- Co-presented project findings at the International Symposium on Electronic Art (ISEA) 2024, highlighting the intersection of robotics, sound, and user experience.
- Co-authored a publication on embodied composition methods for imagining robotic sound space, accepted at the ACM/IEEE HRI 2025 Conference.

### **Project Manager and Education Innovation Consultant**

Erasmus University Rotterdam, Innovation Hub

August 2020 – August 2022

- Within the school of Culture, Communication and History, I assisted with tutorial facilitation, student assessment, and academic guidance.
- Advised course coordinators on how to make their classes more interactive, especially remotely conducted ones.
- Supervised a team of two student assistants to support faculty needs during COVID-19 and beyond.

### **Project Manager Intern**

DPDK Digital Agency, Rotterdam

January 2020 – August 2020

- Oversaw several projects related to internal marketing (e.g. newsletter) and external marketing (e.g. SEO optimisation).
- I redesigned and implemented a new intranet platform for all employees.

### **Teaching Assistant Undergraduate and Graduate Courses**

Erasmus University of Rotterdam

January 2020 – August 2022

- Created lecture videos for Media Technologies and Their Impacts.
- Taught Media Technologies and Their Impacts, Surveillance and Society, Qualitative Methods in the Social Sciences, Advanced Qualitative Methods in the Social Sciences and Emerging Media Technologies.

### **Research Software Skills**

- Atlas.ti
- NVIVO
- Overleaf
- Microsoft Office: PowerPoint, Excel, Word, etc.
- Qualtrics Survey Software

- Mentimeter

## **Presentations**

### **Reconceptualising Human-Robot Collaboration in AEC: A PRISMA Scoping Review**

*43<sup>rd</sup> International Symposium on Automation and Robotics in Construction (ISARC) 2026, National University Singapore, SG*  
June 2026

### **The Invisible Work of Robotic Surgery: How Specialists Support, Shoulder, and Sustain Human-Robot Collaboration**

*ACM DIS 2026, National University Singapore, SG*  
June 2026

### **The Choreography of Care: An Ethnographic Study of Human-Robot Collaboration in Makoplasty Surgeries**

*ACM CHI 2026, Remote Video Presentation*  
April 2026

### **Understanding the Human(s) in Human-Robot Collaboration: A Socio-Technical Inquiry into Robot-Assisted Surgical Work**

*HCC Seminar Series, Aalborg University, DK*  
March 2026

### **The Choreography of Care: An Ethnographic Study of Human-Robot Collaboration in Makoplasty Surgeries**

*QCR Seminar Series, QUT Centre for Robotics, AU*  
March 2026

### **Exploring Human-Robot Collaboration in Surgery: Initial Insights from Mako Product Specialists**

*ACM OzCHI 2025 Conference, Sydney, AU*  
December 2025

### **Removing the Drapes: Reflexive Insights on Research into Real-World Surgical Human-Robot Collaboration**

*ACM OzCHI 2025 Conference, Sydney, AU*  
December 2025

### **Opening the Theatre Doors: Observing Human-Robot Collaboration in Surgical Practice**

*HDR Seminar Series, School of Architecture and Built Environment, QUT*  
November 2025

**Understanding and Designing for the Human(s) in Surgical Human-Robot Collaboration**

*HDR Symposium, Faculty of Engineering, QUT*

October 2025

**PUP116 - Human Factors & Ergonomics**

*Guest Lecture, Faculty of Health, QUT*

September 2025

**Read Them Sideways: Robotically-Assisted Surgery with Jasper Vermeulen**

*QUT Digital Media Research Centre, AU*

June 2025

**Human-Robot Collaboration in Surgical Robotics**

*Technical University Eindhoven, Human-Technology Interaction Group, NL*

May 2025

**Industry 5.0 in Surgery?**

*Erasmus University of Rotterdam, SEISMEC Group, NL*

May 2025

**Exploring Human Performance in Mako-Assisted Hip Replacement Surgeries**

*Chartered Institute of Ergonomics and Human Factors (CIEHF) Annual Conference, UK*

April 2025

**Human-Centred Robotics in Total Arthroplasty Surgeries**

*Queensland Orthopaedic Research Fund (QORF), Annual Meeting, AU*

April 2025

**Investigating human factors in Mako-assisted Total Knee Arthroplasty surgeries**

*ACM/IEEE HRI 2025 Conference, Melbourne, AU*

March 2025

**PhD Research Overview Presentation**

*Australian Computer-Human Interaction Conference 2024, Doctoral Consortium, AU*

December 2024

**Why is Human Factors important in Robotics?**

*Human Factors and Ergonomics Society of Australia (HFESA), Annual Conference, AU*

November 2024

**"Meet Your Mako: Exploring Human-Robot Collaboration in the Operating Room"**

*QUT 3MT Competition, Faculty of Engineering Final, AU*

July 2024

## Leadership & Service

### **Invited: Australian Computer-Human Interaction Conference 2026 Committee Member, Publications Co-Chair 2026 (Present)**

Together with Dr Joel Harman, I am responsible for the publication pipeline of the OzCHI2026 conference.

### **Organiser and Presenter, Socio-Technical Grounded Theory in HRI: A Hands-on Tutorial for Coding and Memoing, IEEE RO-MAN**

*2026 (Present)*

Building on the work of Prof Hoda, I will deliver a tutorial at IEEE RO-MAN 2026 on applying the Socio-Technical Grounded Theory for data analysis.

### **Elected: SIGCHI Committee Member: Industry Co-Chair**

*2025 – Present*

Part of the Special Interest Group for Computer-Human Interaction (SIGCHI) Committee. My role focuses on connecting the Brisbane CHI community with relevant industry professionals, primarily Human Factors professionals and designers.

### **Nominated: Australian Cobotics Centre (ACC) PhD Representative**

*2025–Present*

Represented ACC PhD candidates in centre-level meetings, supporting student engagement and feedback processes.

### **Elected: Higher Degree Research (HDR) Representative, Research Degrees Committee**

*2025–Present*

Advocating for PhD student interests in university research policy discussions and programme development.

### **Organizer and Presenter, Practical Workshop on How to Use Atlas.ti for Qualitative Data Analysis, QUT Digital Media Research Centre Annual Summer School**

*2024–2025, 2025-2026*

Together with PhD Researcher Mal Fituch from the Faculty of Law, I designed and delivered a workshop at the DMRC summer school, teaching participants how to use the Atlas.ti software for qualitative coding in their PhD data analysis.

### **Elected: Chartered Institute of Ergonomics and Human Factors (CIEHF) Annual Meeting (PhD) Student Volunteer**

*2025*

Together with other student volunteers, we supported the organising committee by ensuring all the sessions ran smoothly.

### **Nominated: ACC Symposium Organising Committee Member**

*2025*

Contributed to the planning and execution of the annual ACC symposium, including speaker coordination and event logistics.

Invited: **Organiser and Presenter, Virtual, Augmented, and Mixed Reality for Human-Robot Interaction (VAM-HRI) Workshop, ACM/IEEE HRI 2025 Conference**

*2024–2025*

Led the in-person session of the VAM-HRI workshop at the HRI 2025 Conference in Melbourne, coordinating with international researchers and facilitating discussions on immersive technologies in human-robot interaction.

Elected: **QUT Centre for Robotics (QCR) PhD Representative**

*2024–2025*

Acted as liaison between PhD researchers and QCR leadership, facilitating communication and organising PhD community events.

Elected: **Australian Computer-Human Interaction Conference 2024 (PhD) Student Volunteer**

*2024*

Together with other student volunteers, we supported the organising committee by ensuring all the sessions ran smoothly.

Elected: **Faculty of Engineering Student Representative, Student Representative Council (SRC)**

*2023–2024*

Elected representative advocating for engineering students' interests within the QUT Student Guild, participating in policy discussions and decision-making processes.

## Volunteering

Invited: **FIRST Lego Robotics Competition Judge (Brisbane, Australia)**

*2024 – Present*

The children (6-12 y/o) present their robot project to the other judges and me. We give the children marks and provide some constructive feedback.

## Languages

- Dutch (Native)
- English (Near Native)
- German (Conversational in context)
- Spanish (Basic Proficiency)
- Japanese (Elementary Proficiency)

## Awards

- **Research Achievement Award 2026** (Awarded by QUT Centre for Robotics)
- **HDR Success Spotlight 2026** (Awarded by QUT Faculty of Engineering & Science)
- **DIS Best Paper Honourable Mention Award** (Awarded by ACM DIS2026)
- **CHI Best Paper Honourable Mention Award** (Awarded by ACM CHI2026)
- **Centre Supporter 2025** (Awarded by Australian Cobotics Centre)
- **Contribution to Public Debate 2025** (Awarded by Australian Cobotics Centre)
- **Recognition Award for Seminar Delivery 2025** (Awarded by QUT School of Architecture and Built Environment)
- **HDR Success Spotlight 2025** (Awarded by QUT Faculty of Engineering & Science)
- **Emerging Leader Award 2024** (Awarded by Australian Cobotics Centre)
- **Most Radical Research Funding Pitch Award 2024** (Awarded by Australian Computer Human-Interaction Society)
- **Industry Research Collaboration Award 2024** (Awarded by Australian Cobotics Centre)
- **EPIC Centre Citizen Award 2023** (Awarded by Australian Cobotics Centre)
- **Best Undergraduate Thesis Nominee** (Awarded by Erasmus University of Rotterdam)